

3.6 Carboxylic Acids & Derivatives

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.6 Carboxylic Acids & Derivatives
Difficulty	Easy

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

This question is about carboxylic acids.

State the general formula of a carboxylic acid

[1 mark]

Question 1b

Name the carboxylic acid shown in Fig. 1.1.

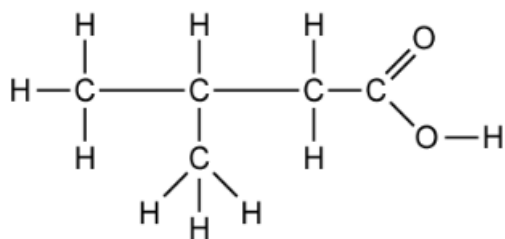


Fig. 1.1

[1 mark]

Question 1c

- i)
Write a balanced symbol equation to show the dissociation of the acid from part (b).
- ii)
State where the position of the equilibrium lies and what this says about the strength of the acid.

[1]

[2]

[3 marks]

Question 1d

Write a balanced symbol equation, including state symbols, for the reaction of propanoic acid with sodium hydrogen carbonate powder to form the soluble sodium propanoate salt and two other products.

[3 marks]**Question 2a**

This question is about esters.

Complete the table to show how some common esters are formed.

Carboxylic Acid	Alcohol	Ester
ethanoic acid		methyl ethanoate
	ethanol	ethyl propanoate
butanoic acid	propan-1-ol	

[3 marks]**Question 2b**

All of the esters in part (a) are formed via a condensation reaction.

State the conditions needed for this reaction.

[2 marks]

Question 2c

Each of the esters in part (a) can be hydrolysed to reform the carboxylic acid and alcohol.

Give the reagents and conditions needed for this reaction.

[2 marks]

Question 2d

Suggest why esters are used in the manufacture of foods.

[1 mark]

Question 3a

This question is about carboxylic acids and esters.

Draw the structures of:

i)
3-methylbutanoic acid

[1]

ii)
methyl ethanoate

[1]

[2 marks]

Question 3b

2-methylbutanoic acid can be produced by the oxidation of 2-methylbutan-1-ol using acidified $\text{K}_2\text{Cr}_2\text{O}_7$.

Give the colour change that would be observed during this reaction.

[1 mark]

Question 3c

Name the products formed when methyl ethanoate is hydrolysed under acidic conditions.

[1 mark]

Question 3d

Methyl ethanoate can also be hydrolysed under alkaline conditions using sodium hydroxide.

i)

Draw the displayed formula of the salt and alcohol formed in this reaction.

ii)

How is the salt converted into carboxylic acid?

iii)

Apart from the conditions, and products made, give one other difference between acid and alkaline hydrolysis.

[4 marks]

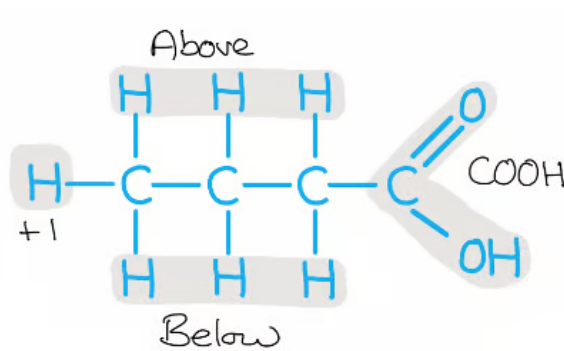
Model Answers: Easy

1a

a) The general formula of a carboxylic acid is:

- $C_nH_{2n+1}COOH$; [1 mark]

[Total: 1 mark]



- For every carbon in the hydrocarbon chain
 - There is a hydrogen above
 - There is a hydrogen below
 - There is also one hydrogen at the end of the chain
 - This means we have $2n+1$ hydrogen for n carbons
 - This is the C_nH_{2n+1} portion of the general formula
- There is also the $COOH$ at the other end of the chain
- So, overall we have $C_nH_{2n+1}COOH$

1b

b) The carboxylic acid shown in Fig.1.1 is:

- 3-methylbutanoic acid; [1 mark]

[Total: 1 mark]

- The longest carbon chain is 4 carbons long
 - Therefore, butan- stem

- There is a carboxylic acid
 - Therefore, butanoic acid
 - **Remember:** You do not have to number the acid as it is automatically carbon-1, as this is the priority group
- There is a methyl group on carbon-3
 - Therefore, 3-methylbutanoic acid

1c

c)

i) The equation to show the dissociation of the acid is:

- $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{COOH} \rightleftharpoons \text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{COO}^- + \text{H}^+$
OR
 $\text{C}_4\text{H}_9\text{COOH} \rightleftharpoons \text{C}_4\text{H}_9\text{COO}^- + \text{H}^+$; [1 mark]

ii) The position of the equilibrium lies:

- To the left; [1 mark]
- (This tells us that it is a) weak acid; [1 mark]

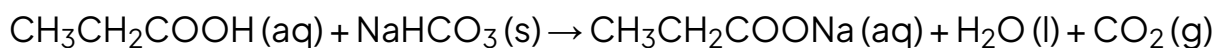
[Total: 3 marks]

- All acids dissociate
 - Strong acids can fully dissociate into their component ions and the equilibrium lies to the right
 - For sulfuric acid this is a two-step process
 - Step 1 is the full dissociation of the sulfuric acid to HSO_4^-
 - $\text{H}_2\text{SO}_4 \rightarrow \text{HSO}_4^- + \text{H}^+$
 - Step 2 is an equilibrium as the HSO_4^- acts as a weak acid
 - $\text{HSO}_4^- \rightleftharpoons \text{H}^+ + \text{SO}_4^{2-}$
 - Carboxylic acids such as CH_3COOH are weak acids – they do **not** fully dissociate and the equilibrium lies to the left

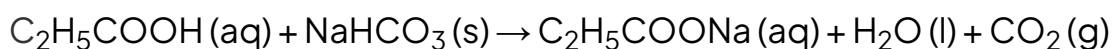


1d

d) The balanced symbol equation, including state symbols, for the reaction of propanoic acid with sodium hydrogen carbonate powder is:



OR



- Correct formulae of reactants; [1 mark]
- Correct formulae of products; [1 mark]
- Correct state symbols; [1 mark]

[Total: 3 marks]

- **Remember:** The reaction of an acid with a carbonate/hydrogen carbonate forms the salt plus water plus carbon dioxide
- You would be expected to know or deduce the chemical formulae and state symbols for all of the chemicals in this reaction

2a

a) The completed table is:

Carboxylic Acid	Alcohol	Ester
ethanoic acid	methanol; [1 mark]	methyl ethanoate
propanoic acid; [1 mark]	ethanol	ethyl propanoate
butanoic acid	propan-1-ol	propyl butanoate; [1 mark]

[Total: 3 marks]

- **Remember:** When naming an ester, the alcohol provides the first part of the name, and the carboxylic acid the second part of the name.

2b

b) The conditions needed to form an ester by reacting an alcohol and carboxylic acid together are:

- Conc sulfuric acid catalyst; [1 mark]
- Heat under reflux; [1 mark]

[2 marks]

- This is also called esterification
- You need to learn the conditions for this reaction

2c

c) The conditions needed for this reaction are:

- Dilute acid / dilute sulfuric acid / dilute hydrochloric acid
OR
Dilute alkali / sodium hydroxide; [1 mark]
- Heat **AND** reflux; [1 mark]

[Total: 2 marks]

- Either the acid or alkali can be used but the reaction with dilute acid is reversible whereas the reaction with you can deduce one is for the reagents and another for the conditions

2d

d) Esters are used in the manufacture of foods because:

- They have nice flavours/ fruity smells; [1 mark]

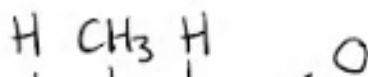
[Total: 1 mark]

- There are many other uses of esters including:
 - Perfumes
 - Glue
 - Nail Varnish
 - Solvents
 - Soaps

3a

a) The structures are:

The structures of 3-methylbutanoic acid and methyl ethanoate are:



2d

d) Esters are used in the manufacture of foods because:

- They have nice flavours/ fruity smells; [1 mark]

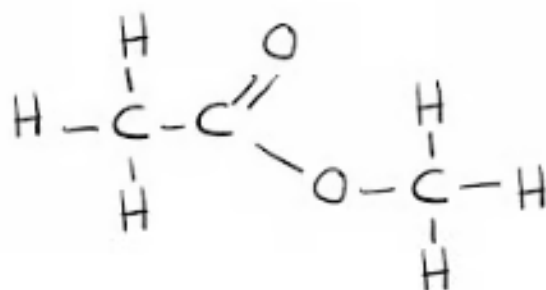
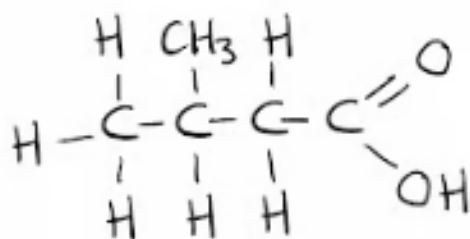
[Total: 1 mark]

- There are many other uses of esters including:
 - Perfumes
 - Glue
 - Nail Varnish
 - Solvents
 - Soaps

3a

a) The structures are:

The structures of 3-methylbutanoic acid and methyl ethanoate are:



- Correct displayed formula of 3-methylbutanoic acid; [1 mark]
- Correct displayed formula of methyl ethanoate; [1 mark]

- Correct displayed formula of methyl ethanoate; [1 mark]

[Total: 2 marks]

- **Remember:** When drawing carboxylic acids, the C of the COOH group is counted as the first carbon, so the methyl group goes on the H second from the left
- When drawing esters, such as methyl ethanoate,
 - The first part of the displayed formula indicates the ethanoate
 - The second part of the displayed formula indicates the methyl

3b

b) The colour change that would occur during this reaction is:

- Orange to green; [1 mark]

[Total: 1 mark]

- The oxidising agent, $K_2Cr_2O_7$ gets reduced causing the solutions to change colour
- The orange dichromate ions ($Cr_2O_7^{2-}$) are reduced to green Cr^{3+} ions

3c

c) The products formed are:

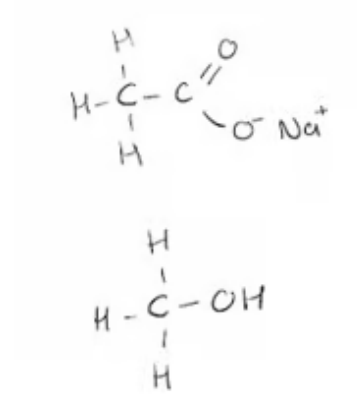
- Methanol
AND
Ethanoic acid; [1 mark]

[Total: 1 mark]

- When an ester is hydrolysed using dilute acid, the original carboxylic acid and alcohol is formed
- **Remember:** For an ester, the first part of the name comes from the alcohol, and the second part from the carboxylic acid

3d

d) i) The salt and alcohol formed during alkaline hydrolysis are:



- Correct displayed formula for sodium ethanoate; [1 mark]
- Correct displayed formula for methanol; [1 mark]

ii) The salt is converted into the carboxylic acid by

- Adding dilute acid / dilute HCl / dilute H₂SO₄; [1 mark]

iii) One other difference between acid and alkaline hydrolysis is:

- An equilibrium is established in acidic hydrolysis

OR

Acid hydrolysis is a reversible reaction; [1 mark]

[Total: 4 marks]

- During acid hydrolysis, an equilibrium is established so there is always some ester present in addition to the carboxylic acid and alcohol



- Alkaline hydrolysis is an irreversible reaction, producing the salt of the carboxylic acid and an alcohol



- The salt is then acidified to produce the carboxylic acid
- Alkaline hydrolysis is much quicker than acidic hydrolysis and is the basis of soap making